

Phosphate plant -- Sensors

Section report: Sensors

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-02-09T23:00:00 (40 days, 888 operating hours)
Product	Dap from 29.0 % P2O5 ore, 165.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 960 hourly intervals of 64 signals generated 2026-09-23T14:50:30.737 basis :screening_plant_model

The inferential sensors: process, vision and acoustics

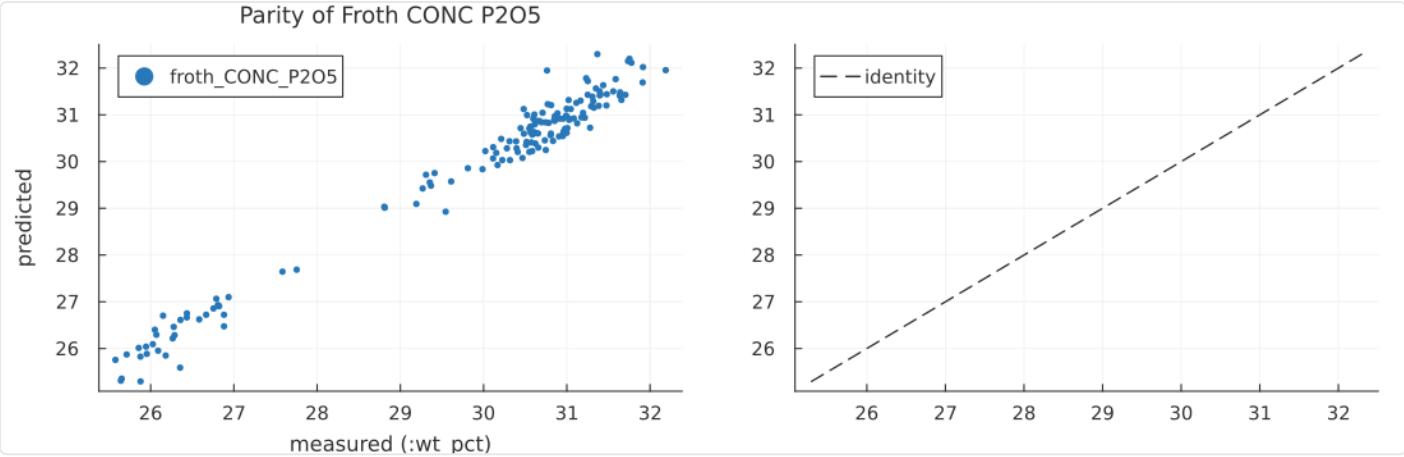
THE REGISTER OF THE INFERENTIAL SENSORS

Quantity	Value									
Sensors	14 inferential models, of which 4 read an image or a sound									
Median R2	0.69358									
Best sensor	:pump_PUMP_VIBRATION									
Status	:at_risk									

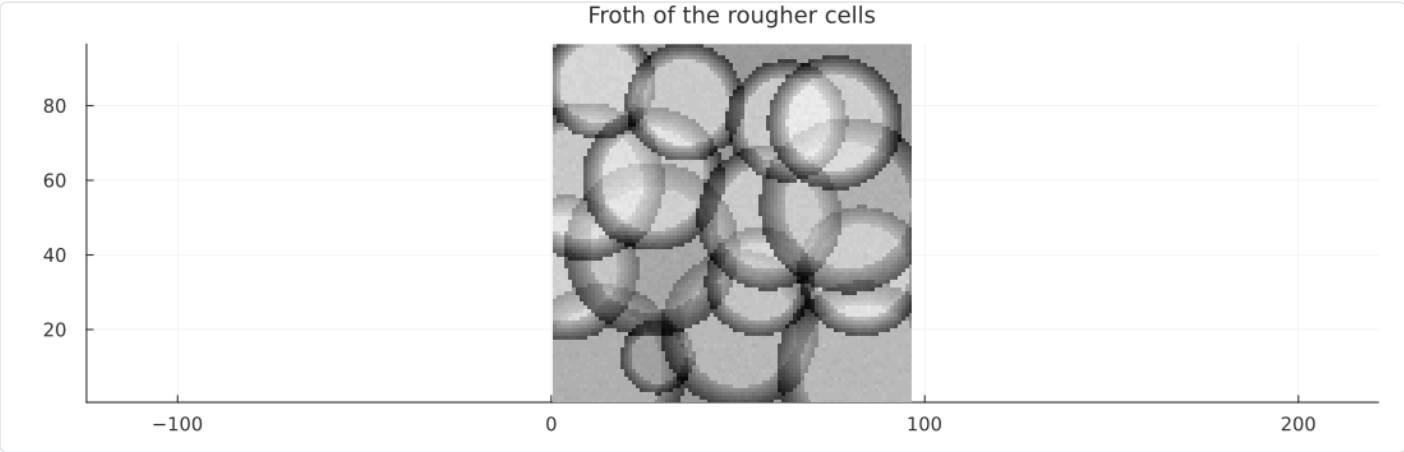
Sensor	Infers	Modality	Unit	R2	RMSE	Bias	In domain %	Drift	Status	Inputs
:pump_PUMP_VIBRATION	:PUMP_VIBRATION	:acoustic	:count	0.98829	0.159723	0.0047	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:froth_CONC_P2O5	:CONC_P2O5	:visual	:wt_pct	0.97283	0.324348	-0.0342	n/a	0	Healthy	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:mill_MILL_POWER	:MILL_POWER	:acoustic	:kw	0.96879	81.1645	-18.88	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:process_CONC_P2O5	:CONC_P2O5	:process	:wt_pct	0.93534	0.475035	-0.0182	100	-0.0073	Healthy	FLOT_FEED_P2O5, CYCLONE_P80, REAGENT_FLOW
:process_STEAM_FLOW	:STEAM_FLOW	:process	:t_ph	0.86722	2.5258	0.1317	100	0.0527	Healthy	EVAP_FEED_FLOW, STRONG_ACID_P2O5, EVAP_VACUUM
:process_MILL_POWER	:MILL_POWER	:process	:kw	0.77055	228.2205	7.62	100	3.04	Healthy	ROM_FEED, SLURRY_DENSITY, MILL_FEED_WATER
:process_CYCLONE_P80	:CYCLONE_P80	:process	:um	0.75736	8.8503	0.4896	99.64	0.1956	Healthy	MILL_POWER, SLURRY_DENSITY, CYCLONE_PRESSURE, MILL_FEED_WATER
:rock_CYCLONE_P80	:CYCLONE_P80	:visual	:um	0.6298	11.2325	-0.0101	n/a	0	Degraded	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:process	:wt_pct	0.48279	0.260584	-0.0229	100	-0.0092	Healthy	SLURRY_SG, ATTACK_TEMP, FREE_SO4, WASH_WATER_FLOW
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:process	:wt_pct	0.15289	0.612708	-0.0751	100	-0.03	Healthy	EVAP_DENSITY, EVAP_TEMP, EVAP_VACUUM
:process_PRODUCT_N	:PRODUCT_N	:process	:wt_pct	0.13863	0.757532	-0.0388	100	-0.0155	Healthy	AMMONIA_FLOW, PRODUCT_FLOW, MELT_DENSITY
:process_FILTER_RATE	:FILTER_RATE	:process	:t_per_m2_d	0.06971	1.6403	0.0066	100	0.0026	Healthy	FILTER_VACUUM, WEAK_ACID_P2O5, SLURRY_SG, GYPSUM_FREE_P2O5
:process_ATTACK_TEMP	:ATTACK_TEMP	:process	:deg_c	-0.00871	16.5475	0.0254	100	0.0102	Healthy	H2SO4_FLOW, ROCK_FEED, RECYCLE_ACID_FLOW, SLURRY_SG
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:process	:wt_pct	-0.01639	0.075369	0.0068	100	0.0027	Healthy	FILTER_RATE, FILTER_VACUUM, WASH_WATER_FLOW, WEAK_ACID_P2O5

WHAT THE SENSORS SAY AGAINST WHAT THE PLANT MEASURED

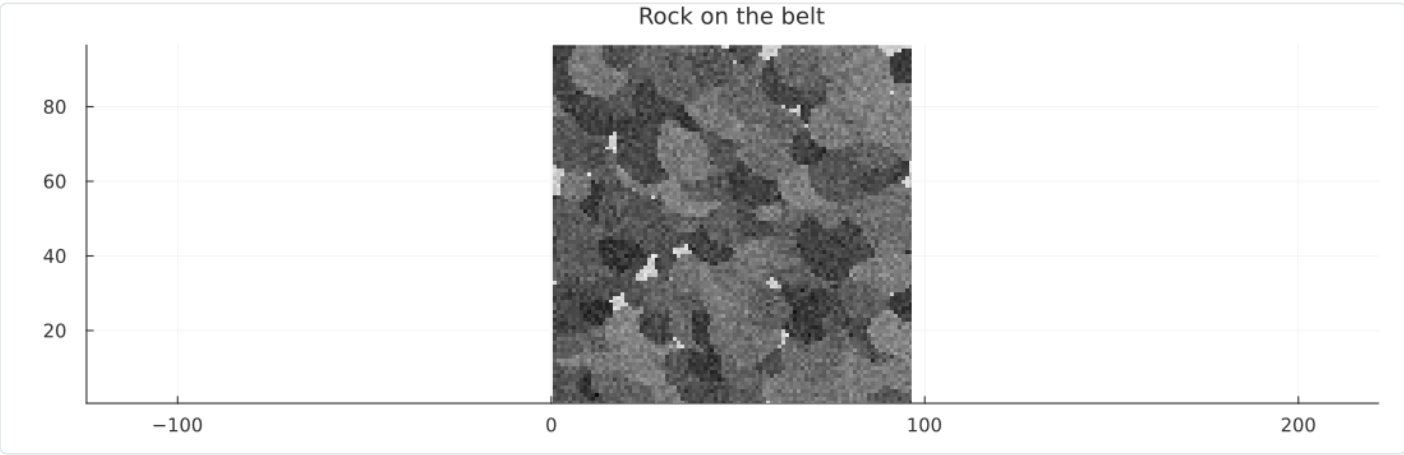
Sensor	Infers	Unit	Estimate	Measured	Difference %	RMSE	Status
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:wt_pct	55.81	56	-0.3355	0.6127	Compliant
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:wt_pct	28.68	28.73	-0.1662	0.2606	Compliant
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:wt_pct	0.4863	0.4726	2.9	0.0754	Compliant
:process_CYCLONE_P80	:CYCLONE_P80	:um	149.57	150.66	-0.7239	8.85	Compliant
:process_MILL_POWER	:MILL_POWER	:kw	5,506.79	5,487.2	0.357	228.22	Compliant
:process_PRODUCT_N	:PRODUCT_N	:wt_pct	18.09	18.17	-0.4135	0.7575	Compliant
:process_FILTER_RATE	:FILTER_RATE	:t_per_m2_d	7.9	8.02	-1.59	1.64	Compliant
:process_CONC_P2O5	:CONC_P2O5	:wt_pct	30.73	30.85	-0.3844	0.475	Compliant
:process_ATTACK_TEMP	:ATTACK_TEMP	:deg_c	78.02	76.27	2.29	16.55	Compliant
:process_STEAM_FLOW	:STEAM_FLOW	:t_ph	47.82	46.61	2.61	2.53	Compliant



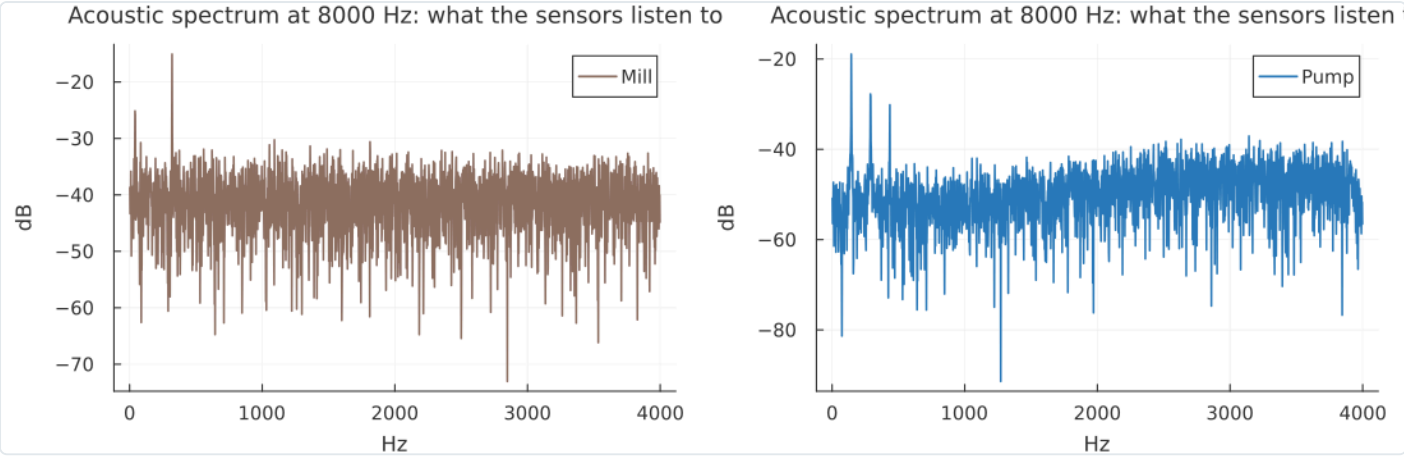
Parity of the froth sensor against the analyser



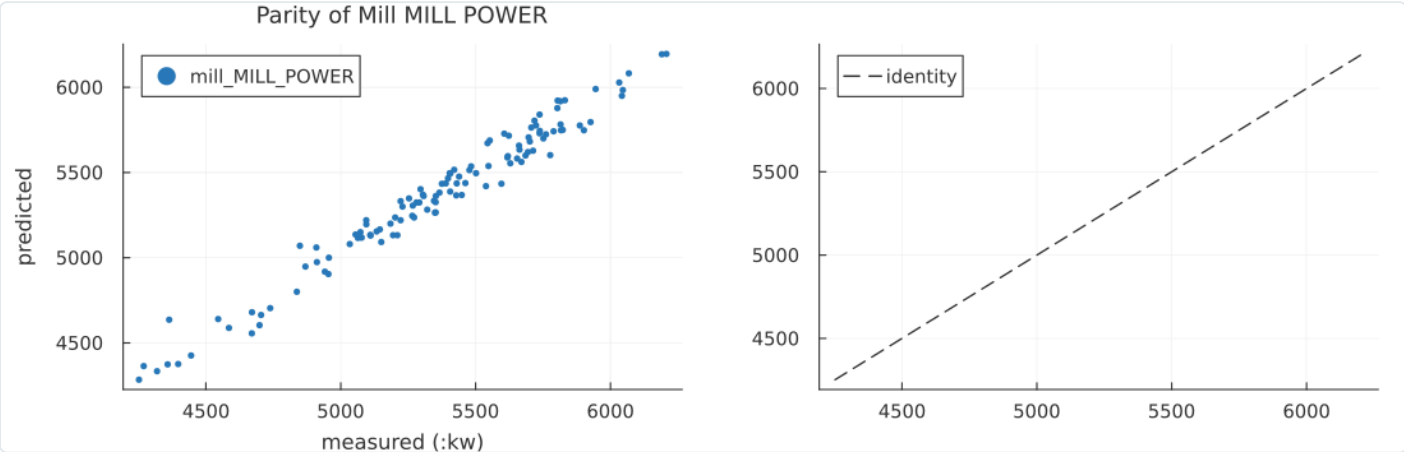
The froth image the visual sensor reads the grade from



The image of the rock the visual sensor reads the grind from



Spectra of the mill and of the pump: the two acoustic channels of the plant



Parity of the acoustic sensor of the mill against the power it draws